

En-Masse MCT Processing Usage Notes:

PRIOR TO RUNNING: ensure you have:

1. THE SCAN FOLDER:

- scan folder name must end with “_#” (e.g. CT_1, CT_2, Scan_5010, this_folder_1234. The number should be unique.
- must contain a subfolder (e.g. “Slices”) with .tiff slices - pass this name in as slicepath. The tif filenames must follow the same naming format and be distinguished by numbers in the name. These must align the scan with the .csv file (either directly or transposed (mirrored across diagonal). THE NUMBERS MUST BE 0-PADDED TO 4 DIGITS FOR LATER PLOTTING TO WORK
- must contain a .txt file (e.g. controls.txt) that specifies how to crop and rotate the scan. Pass the name of this into the python function if it is not controls.txt. The format is:
ang2rot: 206.5 #25
rowrng: [30,260] #anything here is ignored
colrng: [35,265] #likewise here. Just be sure to have a space between this & the important stuff

2. a layout csv file (see example). Ensure it aligns visually with the scan slices (or transposed versions of them). See paper for more details.

RUNNING EX1:

Upon completion of the above, you are now prepared to subsample the scan for use in the segmentation algorithm. Run the following (I have it formatted for a bash script. I have no idea how to format it for Windows):

```
-----  
  
scanfolder='CT_1'  
slice='Slices'
```

```
cfname='controls.txt'
```

```
python3 EX1_subsample.py --folder $scanfolder --slice-path $slice-path  
--controls-fname $cfname
```

If successful, this will produce a file `ct#_new.npz`, where `#` is the scan number.

RUNNING EX2:

copy the `ct1_params.txt` into THE SCAN FOLDER. Change 1 to whatever the scan number is in the filename. the ID'd iso-levels and tier boundaries will be written to this text file.

ensure `ct1_new.npz` is in THE SCAN FOLDER (with 1 again changed to whatever the scan number is in THE SCAN FOLDER's name).

If the images are transposed (to orientation of CSV), pass in `--transpose 1`. else 0.

pass in the number of rows and columns in the scan (e.g. `--nrow 4 --ncol 4`)

pass in the csv path relative to the scan folder, e.g.
`--csv './layout.csv'`

pass in how the slices are named. I use `dddd` to denote where to put 0-padded numbers. The slices must have 4 places of 0-padding. Input e.g.
`--slicename 'TeethScan2_79microns_Test_3_dddd.tif'`
which says for slice 320, the name is
`'TeethScan2_79microns_Test_3_0320.tif'`

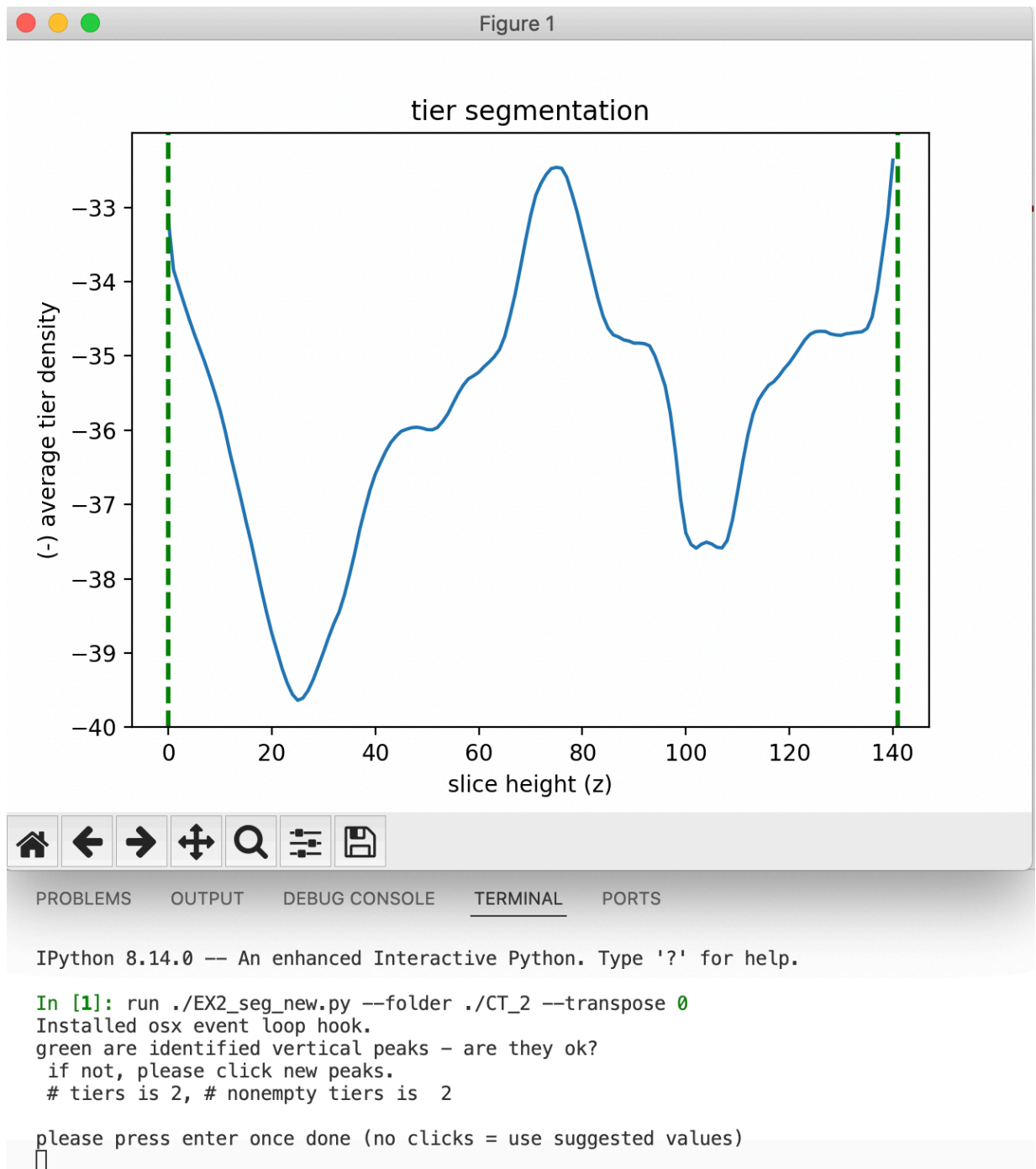
```
scanfolder='CT_1'
```

```
slicesubpath='Slices'  
cfname='controls.txt'  
mirrored=0  
csvpath='../layout.csv'  
nrow=4  
ncol=4  
slcformat='TeethScan2_79microns_Test_3_dddd.tif'
```

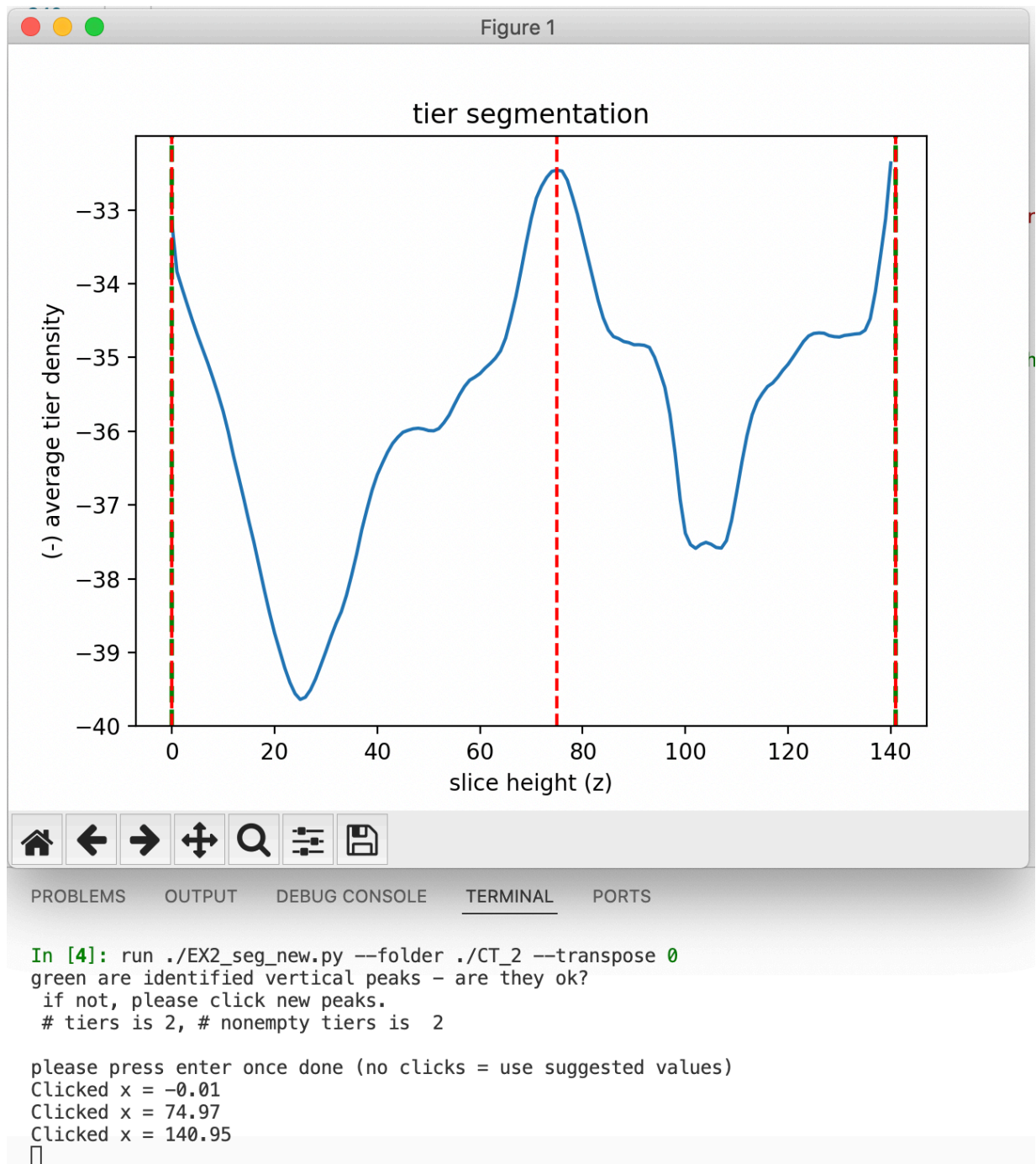
```
python3 EX2_seg_new.py --folder $scanfolder --local_slice_path  
$slicesubpath --transpose $mirrored --nrow $nrow --ncol $ncol --csv  
$csvpath
```

whilst running, you must perform two manual interventions:

1. verifying the tier segmentation. It will prompt you to review the proposed tier boundaries. if the scan has N tiers, there should be N+1 peaks. If any are missing, click all the peaks.

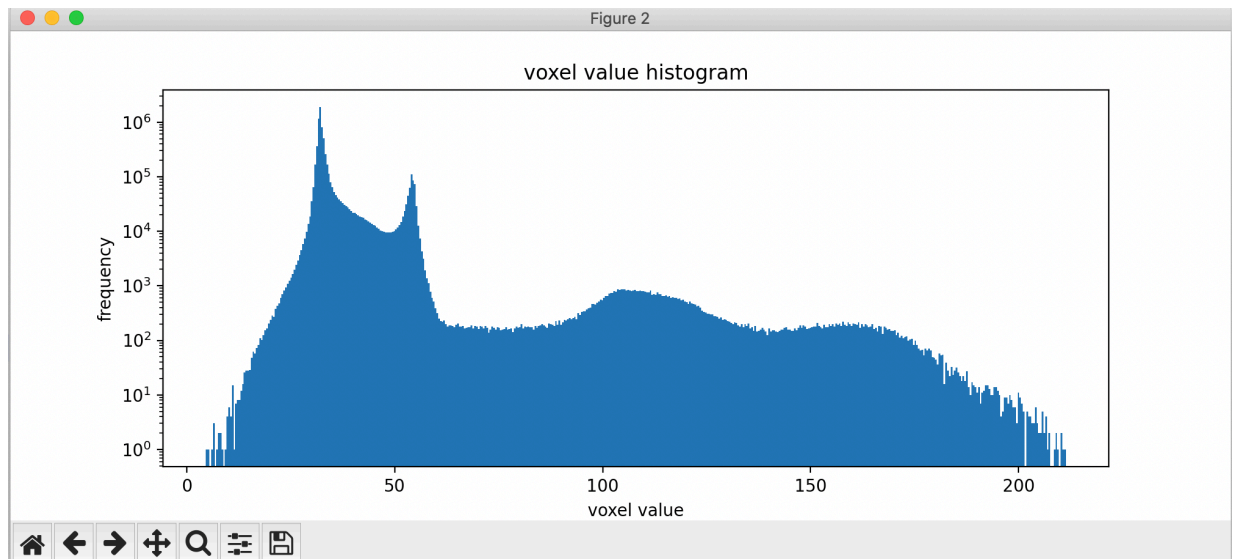


^output before clicking. this scan has two tiers, but it has only identified two peaks (needs 3). If it outputs something that looks correct, press enter.

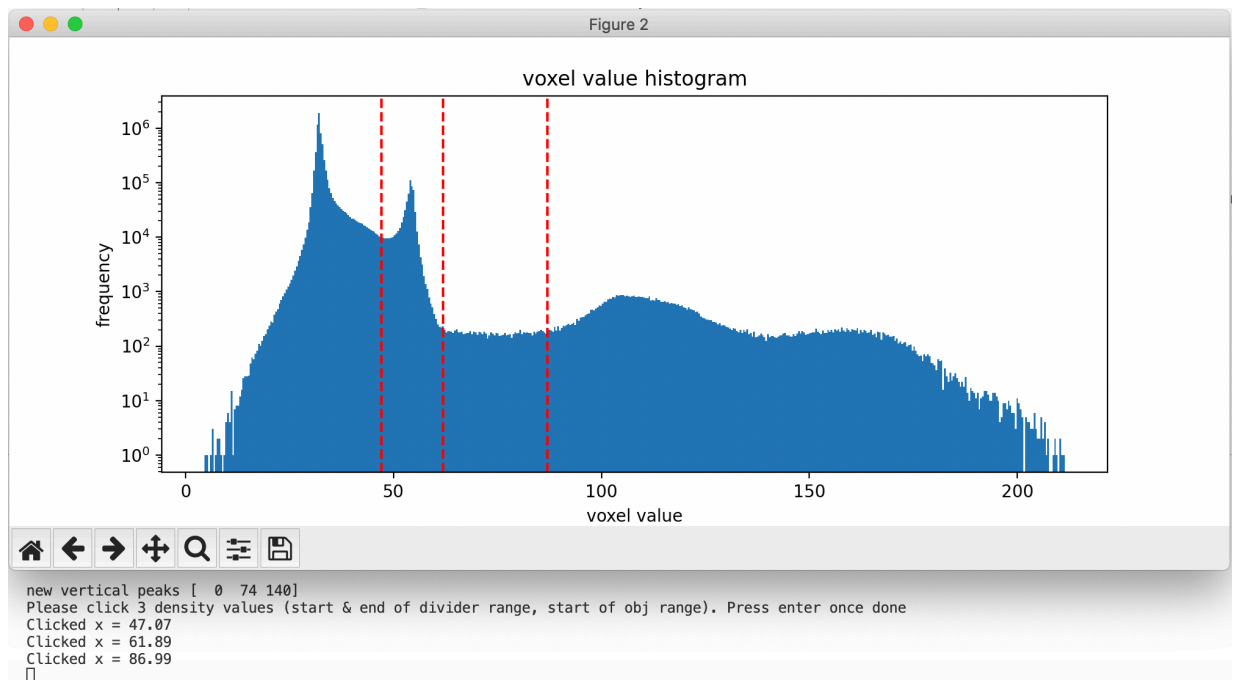


^output after clicking. please click left to right, reclicking any correctly identified peaks. Press enter when done.

2. VOXEL SEGMENTATION. need to identify start and end of divider range, and ~start of object range.



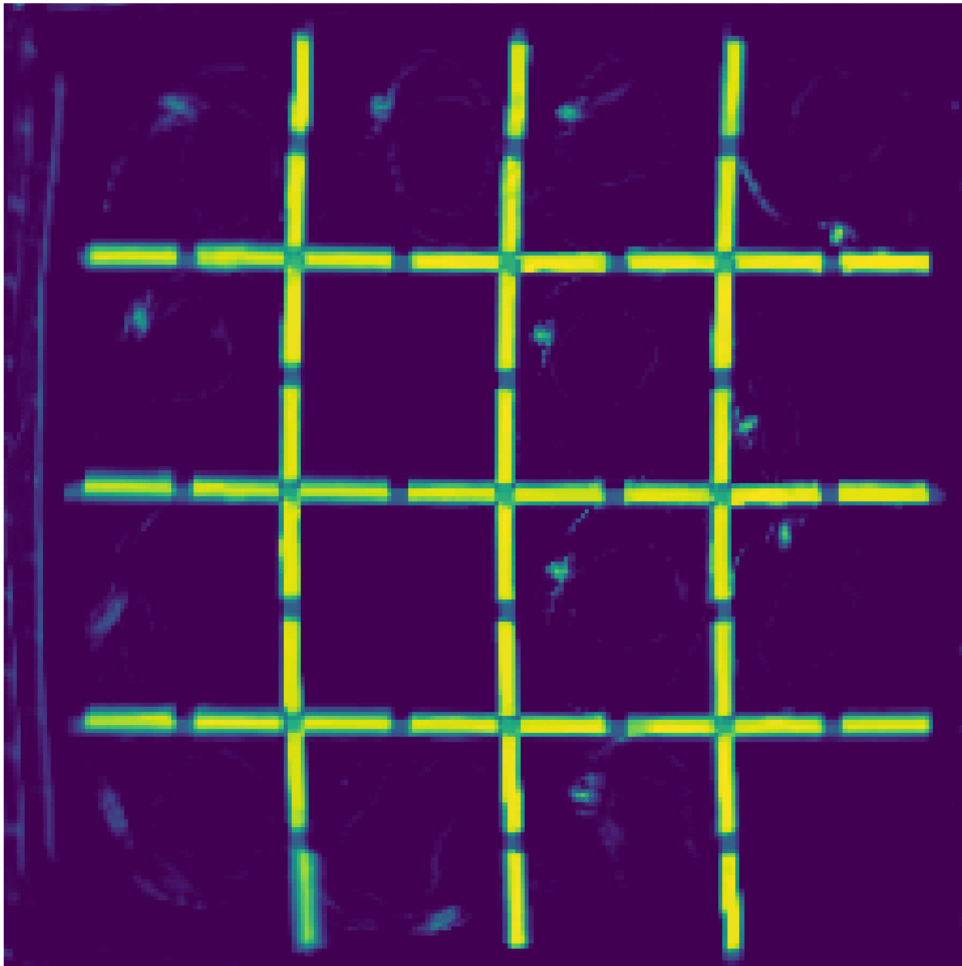
^it will output a figure like this. typically air is the first huge peak, the dividers are the second smaller peak, and the objects are the low third peak.



^click left to right again. Here is what a successful segmentation should roughly look like (start and end of divider range, start of object range). press enter when done.

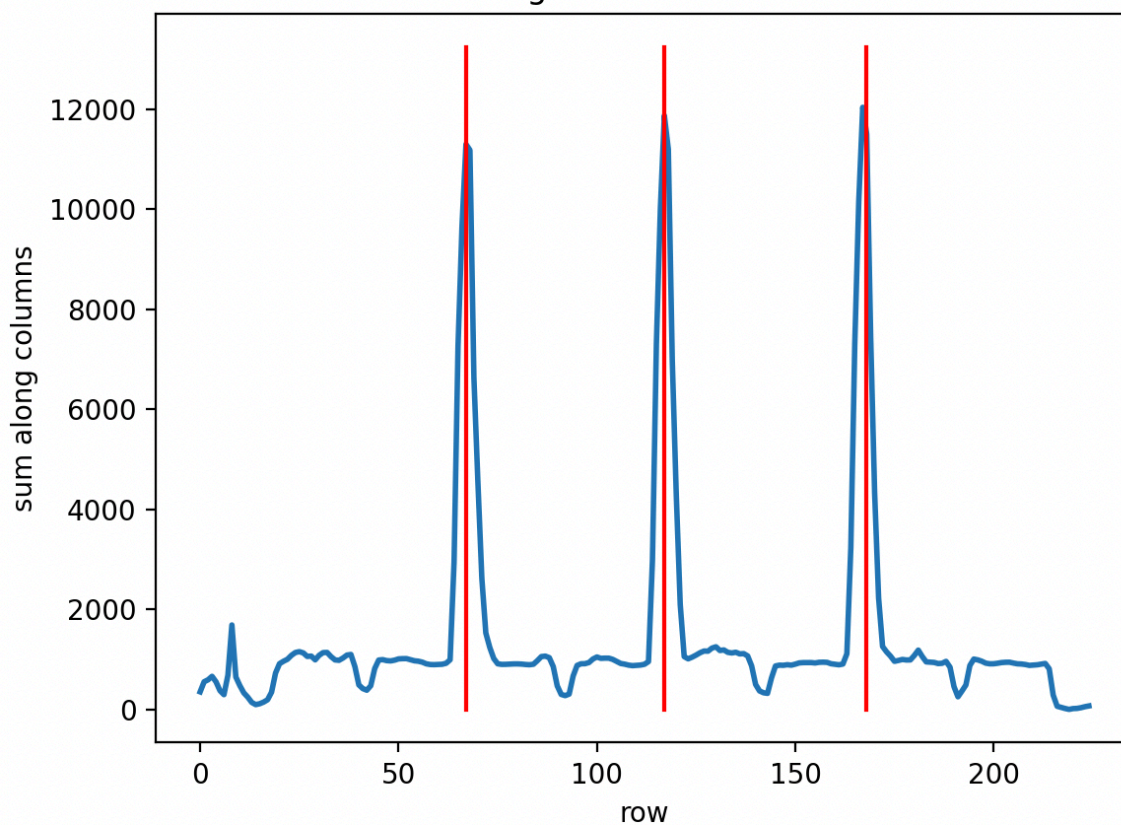
The code will dispense some plots for sanity checks like the following:

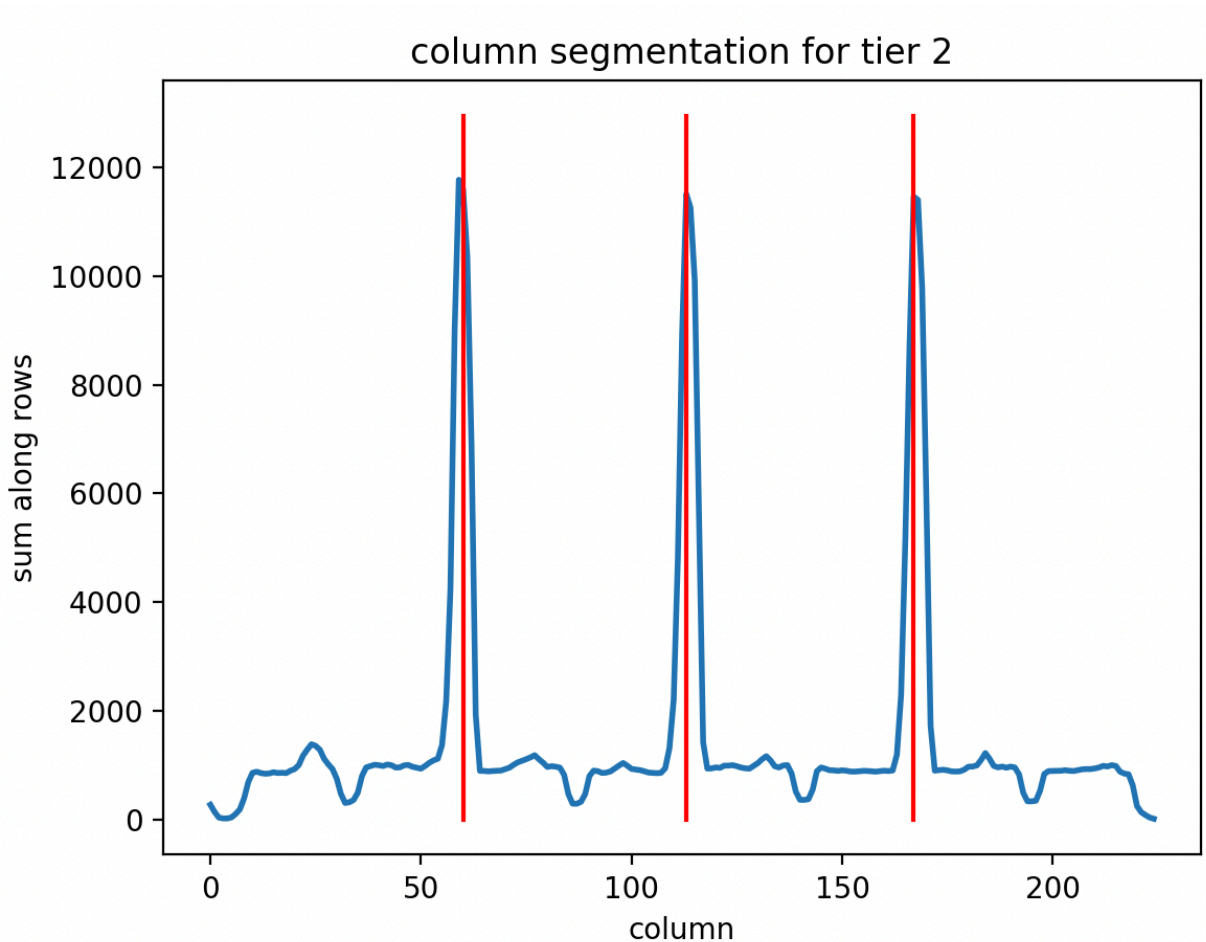
detected dividers for tier 2



^here is an aligned tier boundary mask. It should have the dividers quite horizontal/vertical. It's ok if some of the objects come out, but too many indicates you did not choose the end of the divider change well.

row segmentation for tier 2





if the row/col segmentation plots look very erratic, something in the segmentation has failed.

If you have the slices locally, it will draw the the segmentation on a representative slice image for each tier and show it to you. If it does not line up but the segmentation looks good... email me.

If this has run successfully, it will produce a csv file "CT#.csv" in THE SCAN FOLDER, which specifies how to transform and crop the GIANT scan into the individual objects.

IF you are satisfied with the images, proceed to EX3.

MAKE NOTE OF THE HIGHEST VOXEL VALUE (or find it in the now modified ...params.txt file). YOU NEED THIS FOR EX3.

RUNNING EX3

This code will extract sub-volumes for each object as .npz or .npy files. It is rather slow.

ensure you have CT#.csv in THE SCAN FOLDER. Also ensure you have enough storage available on your machine.

by default, it will put the meshes in the meshsubfolder.

Most importantly: input the third voxel value from EX2 for --iso.

input the resolution scaling (in mm) as res like --res 0.76

If the scan is mirrored, set mirrored=1.

#mandatory inputs:

scanfolder='CT_1'

res=0.76

iso=6000

#optional (if different than default, else delete from function call):

slicesubpath='Slices'

cfname='controls.txt'

mirrored=0

outfolder='Meshes'

```
python3 EX3_chop_n_surf.py --folder $scanfolder --sliceath
$slicesubpath --transpose $mirrored --controls_name $cfname --
meshsubfolder $outfolder --iso $iso --res $res
```

if you get a memory crashout, input e.g. --nworkers 8
(or some number lower than 85% of your CPU count).

If successful, this will output .npz files and .ply files for each surface in the 'Meshes' subfolder of THE SCAN FOLDER.

RUNNING EX3point5

Upon inspecting the output PLY files, it will become apparent if you selected the wrong isolevel. If too low, you get extraneous material that corresponds not to the desired surface. If too high, you've lost part of the desired surface. To this end, I have make a function to resurface the saved .npz volumes and create new surfaces. Please input an isolevel in the goldilocks range, like as follows:

```
-----  
  
#if the meshes are in 'Meshes' subfolder of CT_1,  
#there is no need to input --meshsubfolder Meshes  
  
python3 EX3point5_resurf.py --folder CT_1 --meshsubfolder Meshes --  
iso 3500  
  
-----
```

The .ply files will be overwritten if successful.

once satisfied with the surfaces, you may delete the .npz files, or keep them if you want them for some reason.

RUNNING EX4

If you have dust coming out in your ply's, run this.

If you have multiple disjoint objects in a single cell of the scan... do not run this, it will remove all but one of them.

```
-----  
  
#Meshes and Clean_Meshes are default subfolders of CT_1
```

#if content with that, just run

#python3 EX4_cleanmeshes.py --folder CT_1

#else specify as

python3 EX4_cleanmeshes.py --folder CT_1 --meshsubfolder Meshes --
newmeshsubfolder Clean_Meshes
